

Artificial Intelligence

Outline

1. Introduction
2. Problems and Search
3. Knowledge Representation
4. Advanced Topics
 - Game Playing
 - Uncertainty and Imprecision
 - Planning
 - Machine Learning

References

- **A Modern Approach Artificial Intelligence**
Russell, Norvig. Second Ed, Pearson Education.
- **Artificial Intelligence (1991)**
Elaine Rich & Kevin Knight, Second Ed, Tata McGraw Hill

Introduction

- What is AI?
- The foundations of AI
- A brief history of AI
- The state of the art
- Introductory problems

What is AI?

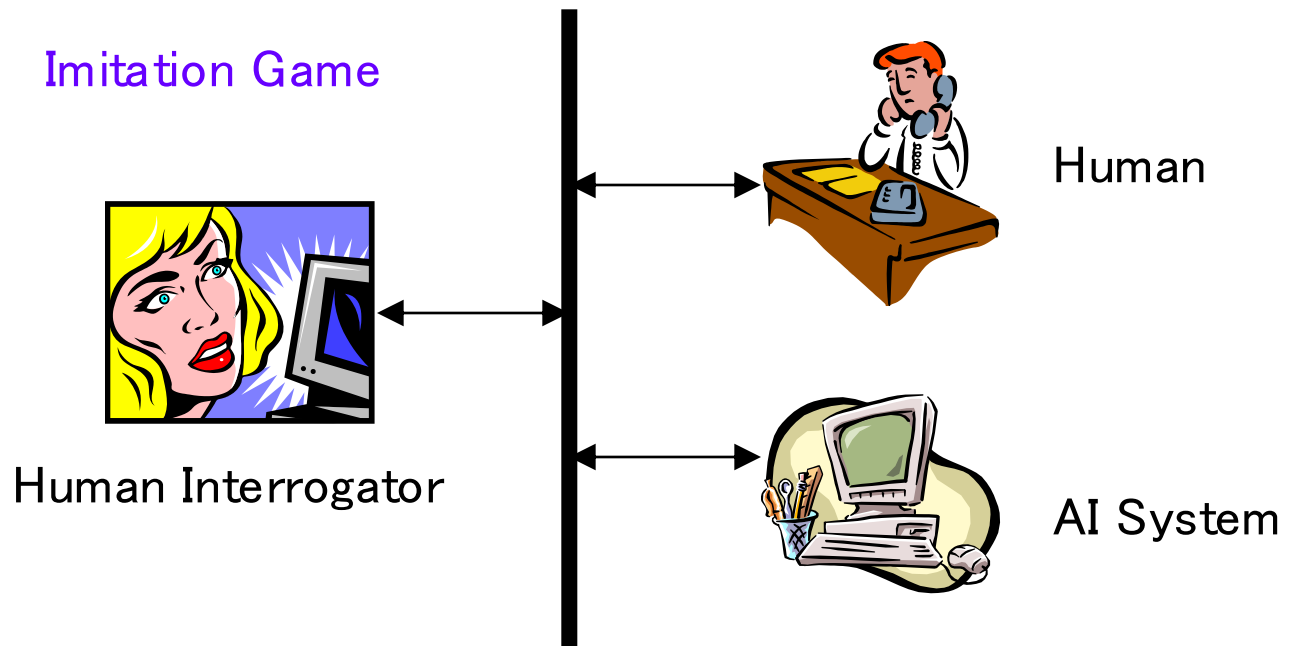
- **Intelligence**: “ability to learn, understand and think” (Oxford dictionary)
- AI is the study of how to make computers make things which at the moment people do better.
- Examples: Speech recognition, Smell, Face, Object, Intuition, Inference, Learning new skills, Decision making, Abstract thinking

What is AI?

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

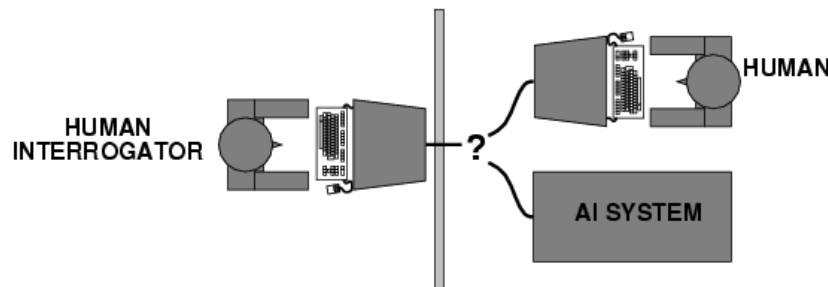
Acting Humanly: The Turing Test

- Alan Turing (1912–1954)
- “Computing Machinery and Intelligence” (1950)



Turing Test...Cont...

- Turing (1950) "Computing machinery and intelligence":
- "Can machines think?" → "Can machines behave intelligently?"
- Operational test for intelligent behavior: the Imitation Game



- Predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes
- Anticipated all major arguments against AI in following 50 years
- Suggested major components of AI: knowledge, reasoning, language understanding, learning
-

Thinking Humanly: Cognitive Modelling

- Not content to have a program correctly solving a problem.

More concerned with comparing its reasoning steps to traces of human solving the same problem.

- Requires testable theories of the workings of the human mind: **cognitive science**.

Thinking Rationally: Laws of Thought

- Aristotle was one of the first to attempt to codify “right thinking”, i.e., irrefutable reasoning processes.
- Formal logic provides a precise notation and rules for representing and reasoning with all kinds of things in the world.
- Obstacles:
 - Informal knowledge representation.
 - Computational complexity and resources.

Acting Rationally

- Acting so as to achieve one's goals, given one's beliefs.
- Does not necessarily involve thinking.
- Advantages:
 - More general than the “laws of thought” approach.
 - More amenable to scientific development than human-based approaches.

The Foundations of AI

- Philosophy (423 BC – present):
 - Logic, methods of reasoning.
 - Mind as a physical system.
 - ☐ – Foundations of learning, language, and rationality
 - ☐ Aristotle – first to formulate laws governing rationality.
- Mathematics (c.800 – present):
 - Formal representation and proof.
 - Algorithms, computation, decidability, tractability.
 - ☐ – Probability, uncertainty handling, fuzzy logic
- Economics
 - Decision theory, game playing, operation research

The Foundations of AI

- Psychology (1879 – present):
 - Adaptation.
 - Phenomena of perception and motor control.
 - ☐ – Experimental techniques.
 - ☐ Cognitive Psychology– brain as information processing machine
- Linguistics (1957 – present):
 - Knowledge representation.
 - ☐ – Grammar
 - ☐ Pattern recognition
 - ☐ .

- Neuro Science – How do brain process information – Brain computer interface
- Neural Network – Graph based neural network structure that can be trained to learn patterns
- Computer engineering – First computer Eniac –1943 , Now –GPU, Quantum Computing

A Brief History of AI

- The gestation of AI (1943 – 1956):
 - 1943: McCulloch & Pitts: Boolean circuit model of brain.
 - 1950: Turing’ s “Computing Machinery and Intelligence”.
 - 1956: McCarthy’ s name “Artificial Intelligence” adopted, LISP invented
- Early enthusiasm, great expectations (1952 – 1969):
 - Early successful AI programs: Samuel’ s checkers, Newell & Simon’ s Logic Theorist, Gelernter’ s Geometry Theorem Prover.
 - Robinson’ s complete algorithm for logical reasoning.

A Brief History of AI

- A dose of reality (1966 – 1974) – First Winter
 - AI discovered computational complexity.
 - Neural network research on perceptron almost disappeared after Minsky & Papert's book in 1969

Reason of failure

based on informed introspection of how human performed task rather than careful analysis of task and deliverable solution

Intractability , combinatorial explosion

Limited resource availability

A Brief History of AI

- Knowledge-based systems (1969 – 1979):
 - – 1969: DENDRAL by Buchanan et al. – molecular structure prediction
 - – 1976: MYCIN by Shortliffe. – diagnosing blood infection, reasoning under uncertainty – how doctors assess impact of evidence on diagnosis
 - – 1979: PROSPECTOR by Duda et al. – mineral extraction
- AI becomes an industry (1980 – 1988):
 - – Expert systems industry booms.
 - – 1981: Japan's 10-year Fifth Generation project

A Brief History of AI

- The return of NNs and novel AI (1986 – present):
 - Mid 80' s: Back-propagation learning algorithm reinvented.
 - – Expert systems industry busts.
 - Connectionst model (NN) competed with symbolic model given by Newel and logistic approach by Mc Carthy

—

A Brief History of AI

- AI Winter – II – Companies fell as they were unable to fulfill promises
- Costly system, specialized hardware
- Difficult to manage and maintain expert system
- Failed in complex domains , too domain specific
- Funds not released

Advancements leading to modern AI

- 1986– Return of Neural Network
- 1988: Resurgence of probability.
- 1988: Novel AI (ALife, GAs, Soft Computing, ...).
- 1995: Agents everywhere.
- 2003: Human-level AI back on the agenda
- 2001 : Big data
- 2013 : Deep Learning
- : Quantum Computing

Risks of AI

- Lethal Autonomous Weapons – weapons that can locate, select and eliminate target without human interventions
- Surveillance and Persuasion– expensive , tedious, legally questionable
- Biased decision
- Impact on employment
- Safety critical application
- Cybersecurity

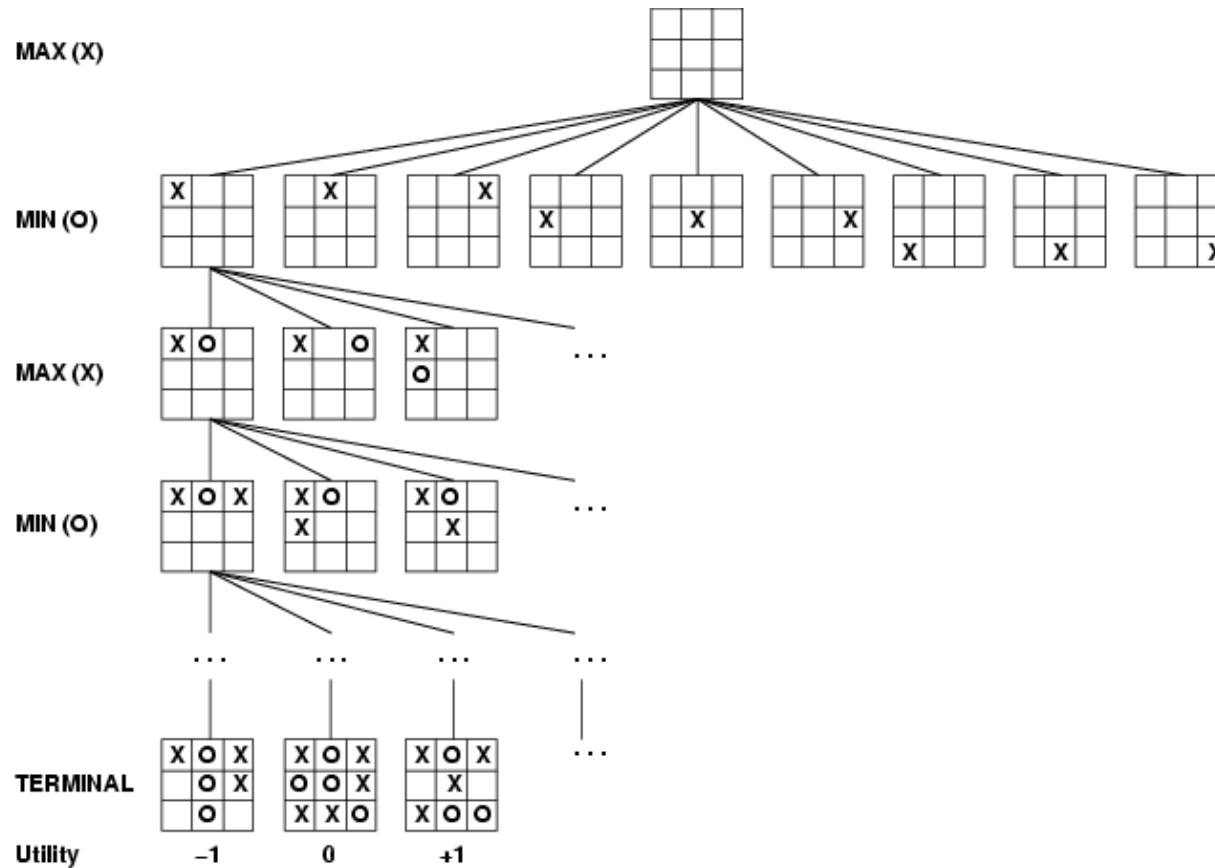
Task Domains of AI

- Mundane Tasks:
 - Perception
 - Vision
 - Speech
 - Natural Languages
 - Understanding
 - Generation
 - Translation
 - Common sense reasoning
 - Robot Control
- Formal Tasks
 - Games : chess, checkers etc
 - Mathematics: Geometry, logic, Proving properties of programs
- Expert Tasks:
 - Engineering (Design, Fault finding, Manufacturing planning)
 - Scientific Analysis
 - Medical Diagnosis
 - Financial Analysis

AI Technique

- Intelligence requires Knowledge
- Knowledge possesses less desirable properties such as:
 - Voluminous
 - Hard to characterize accurately
 - Constantly changing
 - Differs from data that can be used
- AI technique is a method that exploits knowledge that should be represented in such a way that
 - Knowledge captures generalization
 - It can be understood by people who must provide it
 - It can be easily modified to correct errors.
 - It can be used in variety of situations

Game tree (2-player, deterministic)



The State of the Art

- Computer beats human in a chess game.
- Computer–human conversation using speech recognition.
- Expert system controls a spacecraft
- Robot can walk on stairs and hold a cup of water.
- Language translation for webpages.
- Home appliances use fuzzy logic.
- Recommendation systems
-